AJAY GANESAN IYER

Robotics | Motion Planning | State Estimation & Control

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EDUCATION

Texas A&M University, College Station, TX Aug 2024 - Present
Ph.D., Mechanical Engineering

Indian Institute of Technology, Roorkee Jun 2021
M.Tech., Systems and Control

National Institute of Technology, Tiruchirappalli May 2019
B.Tech., Instrumentation and Control

RESEARCH EXPERIENCE

Cooperative Exploration and Target Search Aug 2026 - Present
Research Assistant, Texas A&M University | Advisor: Dr. Srikanth Saripalli

- Developed a **multi-robot exploration system** for one Husky and two Jackals, integrating shared LiDAR mapping and **Extended Kalman Filter (EKF)** localization through odometry-IMU fusion.
- Implemented **Wavefront Frontier Detection (WFD)** and balanced **Hungarian assignment** to distribute search regions using travel-cost estimates and robot task capacities, with reassignment after navigation failures.
- Combined cost-aware **A* planning** with **Model Predictive Path Integral (MPPI) control** to connect team-level assignments to local robot motion; tested goal handling, cancellation, and mission completion.
- Integrated pretrained **YOLO detection with PyTorch/CUDA** and successive-frame confirmation for visual target search. Investigating semantic viewpoint selection and capability-aware allocation to guide where the team searches next.

Perception and Planning for Autonomous Driving Aug 2025 - Jul 2026
Graduate Research Assistant, Texas A&M University | Advisor: Dr. Reza Langari

- Developed a **Frenet-frame trajectory planner** for marked and unmarked gravel roads, generating lateral offsets and speed profiles and selecting feasible trajectories through curvature constraints and weighted costs.
- Tuned **CLRerNet lane detection** and **Segment Anything Model (SAM)** drivable-area segmentation; aligned camera, LiDAR, and odometry data to supply consistent road-boundary estimates to the planner.
- Evaluated planning under glare, low light, dust, sensor dropout, and timing jitter using recorded data and simulation; refined perception thresholds and temporal smoothing to maintain lane continuity.
- Analyzed OpenDriveVLA visual features using **Principal Component Analysis (PCA)** across **6,019 nuScenes samples**. About **10 components retained 80% of scene-token and 97% of map-token variance**, identifying redundancy for token-reduction studies.

Robotic Welding Automation Aug 2024 - Aug 2025
Graduate Research Assistant, Texas A&M University | Advisor: Dr. Prabhakar Pagilla

- Registered CAD geometry to physical workpieces using **three torch-probed points, Kabsch alignment, and Iterative Closest Point (ICP)** refinement, enabling weld-path generation without external metrology.
- Combined **Newton-Raphson inverse kinematics and RRT* planning** with wrist-limit handling and angle unwrapping; timed trajectories for approximately constant torch-tip speed and continuous joint motion.
- Built a **Python/PyQt5/ROS interface** for a UR10e and Miller pulsed-MIG welder, synchronizing robot motion with arc voltage, wire feed, and weld triggering through Arduino PWM/relay control.
- Integrated **infrared weld-pool monitoring** with an Optris PI 05M camera and used bead-height and calibrated temperature feedback to regulate torch travel speed and arc voltage.

TECHNICAL SKILLS

Programming & tools: C++, Python, C/Embedded C, MATLAB/Simulink, ROS 1/2, Linux, Git, LabVIEW
Planning & control: A*, RRT*, MPC, MPPI, Hungarian assignment, sliding-mode control, PID, system identification
Estimation & perception: EKF, SLAM, PCA, RANSAC, camera/LiDAR alignment, OpenCV, PyTorch/CUDA
Robotics & hardware: Husky, Jackal, UR10e, Gazebo, RViz, CARLA, Arduino, Raspberry Pi, infrared thermography

INDUSTRY EXPERIENCE

Robotics Engineer | MIKO Jun 2021 - Jun 2023

Mumbai, India

- Developed **motion-planning and feedback-control software** for expressive head articulation, using system identification, stability testing, and robust controller synthesis to shape the mechanism's response.
- Integrated sensors and tuned feedback controllers for a **self-balancing robot**, taking the control workflow through modeling, implementation, and hardware testing.
- Connected **audio direction-of-arrival estimates to yaw control** so the robot could turn toward an active speaker; implemented remote-control functionality for the mobile platform.

SELECTED PROJECTS & INTERNSHIP

Precision Vehicle Tracking and Disturbance Recovery 2026

Independent project | Predictive control and nonlinear simulation

- Formulated a reference-scheduled **LPV-MPC** controller using a four-state kinematic bicycle model and moving-frame tracking errors to recover from initial offsets and temporary disturbances.
- Assembled a sparse **OSQP** problem over a **20-step horizon**, enforcing steering, acceleration, command-rate, and speed limits at a 100 ms update interval.
- Evaluated three trajectories with **1,520 successful QP solves** and **0.289-0.396 m** full-run position RMSE; analyzed disturbance recovery and command smoothness.

Robust Control for Urban Vehicle Platooning Aug 2020 - Apr 2021

M.Tech. thesis, IIT Roorkee

- Designed longitudinal and lateral control for a **three-vehicle platoon at 8 m/s**, using nonlinear bicycle dynamics and a Luenberger observer for yaw-rate estimation.
- Compared PID, conventional sliding-mode control, **Power Rate Exponential Reaching Law (PREL)-based SMC**, and super-twisting control through integral tracking errors and control energy.
- **PREL-based control achieved the lowest error indices** in the comparison; super-twisting gave lateral convergence without overshoot and the lowest longitudinal control energy.

Ultrasound Needle-Tip Tracking Sep - Dec 2019

Graduate project, IIT Roorkee

- Combined **Random Sample Consensus (RANSAC)** line fitting with a **four-state Kalman filter** in LabVIEW to track needle-tip position when ultrasound speckle obscured image features.
- Reduced position RMSE from **0.79 to 0.03 cm** in artificial tissue and **0.84 to 0.21 cm** in biological tissue versus vision-only tracking, using an NDI-Aurora reference at insertion depths up to 6.5 cm.

Cooperative Light-Source Search Jan - May 2019

B.Tech. thesis, NIT Tiruchirappalli

- Implemented **Particle Swarm Optimization (PSO)** on three LEGO NXT robots, combining personal and swarm-best light measurements with overhead ArUco tracking and ultrasonic obstacle avoidance.
- Demonstrated convergence to the brightest region; one documented hardware trial required **11 iterations**. The related paper received a **Best Paper Award at ICECA 2019**.

EKF-SLAM and Embedded Robot Navigation Jun - Jul 2017

Summer intern, IIT Bombay | Joint project with Alden J. Joseph

- Adapted a MATLAB **EKF-SLAM simulator** for joint pose and landmark estimation, using Jacobian linearization and **Mahalanobis-gated data association**; observed reduced landmark uncertainty during waypoint revisits.
- Extended the simulator with reactive steering and separately programmed a **SPARK-V robot in Embedded C**, combining ultrasonic obstacle avoidance and encoder feedback for room traversal.

PUBLICATIONS & RECOGNITION

- A. G. Iyer, J. Samantaray, S. Ghosh, A. Dey, S. Chakrabarty. *Sliding Mode Control Using Power Rate Exponential Reaching Law for Urban Platooning.* ACODS, 2022.
- M. Adharsh, Ajay G. Iyer, Abhinav G. Athrey, Ashwin V., Venkatesh V. G. *Particle Swarm Optimization Algorithm for Finding the Brightest Spot in an Arena.* IEEE ICECA, 2019. **Best Paper Award.**